

Exercice 2
Advanced Computer Architecture

Anton Schelfhout, Yari Vanherpe

3 maart 2024



Instructor: A. Marinšek

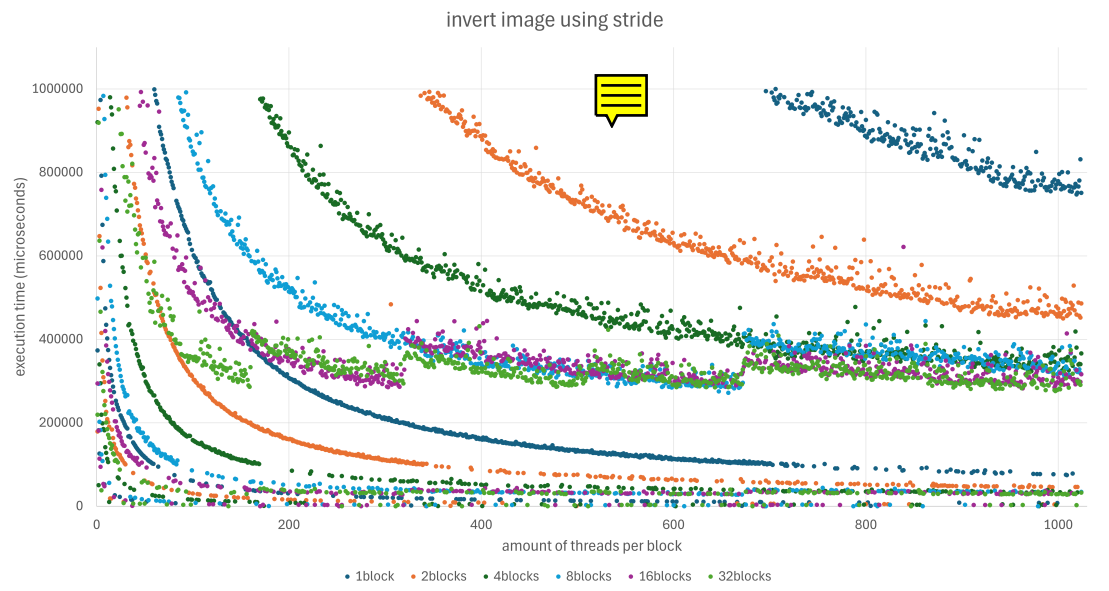
1 Introduction

In this lab session we looked at optimization techniques. We tried to invert the color of images using threads. We also observed what would happen if we make some operations longer than others.

2 Method

First we wrote a function that inverts the color of each pixel. The most basic approach did not use striding.

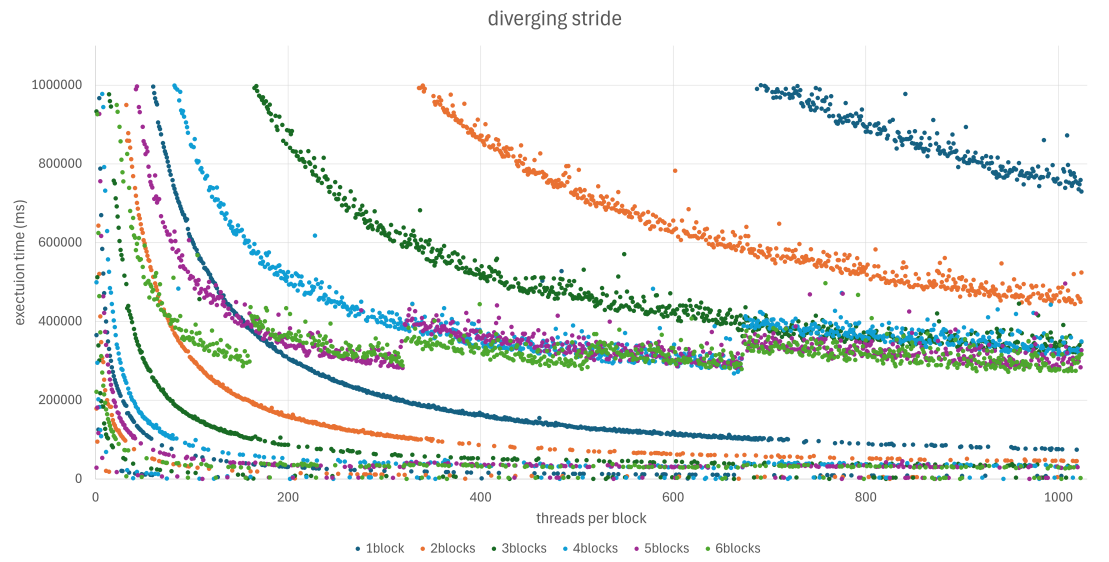
In our second implementation we integrated striding, this means each thread will be used multiple times. By doing this we need less scheduling iterations which should make the code run faster (compared to using each thread once which results in much more blocks). The timing results of this approach are shown in figure 1.



Figuur 1: timing for inverting the image

In figure 1 you can see multiple separate lines for every amount of blocks, this is caused by these tests being blocked by other activity on the GPU.

Lastly we also looked at what would happen if one operation would take much longer than the others, to achieve this we did a complicated operation on the red value of the pixel. Because of this, roughly one third of the threads in a warp are work slower than the others in the same warp, which means they will be idle while the slow threads finish their work. The result of this approach is given in figure 2. A better approach is to divide the work so that all the threads handling the red pixels are part of the same warp or warps (if more than 32 threads for this task). This means not as much cores will be idle and the other threads can continue working.



Figuur 2: timing for diversion

in figure 2 you can again see multiple separate lines for the same reason we got this in figure 1.

3 Observations

From figure 1 we can see that in order to increase performance we need to use more threads. Using more threads can be done by increasing the number of blocks or the number of threads per block. From our timing results you can see that some threads per block do better than others, this falls in line with the size of a warp. Thus the number of threads per block is best a multiple of a warp size.

When using striding the program runs faster than always using new threads, this makes sense because less scheduling itrations are needed.
To even further increase the performance of the second exercise (the case with the longer operation for the red pixel), it seems better to orden the array in a way that all the read values are grouped together. By doing this the threads performing operations on the red values should be in the same warp, which leads to less idle cores.